

DRAFT: The Scalar-Angular-Torsion Framework: II. The Substrate and Universal Anchors

Nathan McKnight
Independent Researcher
(Dated: March 11, 2026)

We define the discretized substrate of the Scalar-Angular-Torsion (SAT) framework, characterized by the 24-cell Hypersphere Unit Cell Vertex (HSUCV) lattice. This section establishes the universal scale anchors—length (l_f), mass (m_0), and expansion rate (c)—which enforce a zero-parameter economy through failure rigidity. We further derive the Projection Constant $B \approx 0.2387$ rad as the fundamental geometric fingerprint governing 4D-to-3D projection distortion and mass scaling.

LAY OVERVIEW

The universe in the SAT framework is not an empty void but is built upon a rigid, four-dimensional “scaffolding” called the **24-cell HSUCV lattice**. Think of this as the master grid for all of reality, a discretized 4D hypersphere that defines where and how filaments can move. Instead of relying on dozens of independent, manually tuned numbers to describe physics, the SAT framework utilizes a **“Single-Anchor Scale”**. By setting just one physical length—the width of a single filament ($l_f \approx 0.7937$ fm)—every other fundamental constant is automatically determined by the geometry of the lattice.

This section also introduces the **Projection Constant (B)**, which is approximately 0.2387 radians. This number is the “geometric fingerprint” of our 4D universe as it projects into 3D space. It acts like a universal gear ratio, determining the “Projective Resistance” (mass) and the thresholds for relativistic effects. Because all constants are tethered to this single, rigid lattice, the framework possesses **“Failure Rigidity”**: the scale cannot be adjusted even slightly without the entire network of physical laws—from the mass of a proton to the strength of gravity—drifting into immediate structural collapse.

I. THE 24-CELL HSUCV LATTICE

The vacuum substrate of the Zottenwelt is modeled as a discretized 4D hypersphere (S^3) expanding at a constant radial rate. The geometry of this expansion is constrained by the 24-cell Hypersphere Unit Cell Vertex (HSUCV) lattice, which serves as the universal frame of reference. This lattice provides the “topological grammar” for all physical interactions, ensuring that worldline filaments satisfy periodicity conditions without mathematical singularities.

As filaments traverse this substrate, their geodesic paths are constrained by the discrete nodes of the 24-cell lattice. This discretization implies that worldline motion is not perfectly smooth but involves mandatory structural “snaps” as filaments maintain topological stability across the HSUCV nodes.

II. UNIVERSAL SCALE ANCHORS

The SAT framework operates within a Zero-Parameter Economy where all physical units are derived from a single anchor pinned to empirical measurement.

A. The Filament Scale l_f

The primary measuring rod of the framework is the Filament Scale l_f , defined as the transverse scale derived from the topological saturation limit. Pinning l_f to the Rydberg Constant yields the universal length:

$$l_f \approx 0.7937 \text{ fm} \quad (1)$$

This single anchor sets the scale for all interrelated physical sectors.

B. The Universal Mass Anchor m_0

The bare mass scale is derived from intrinsic filament tension and the l_f scale. It serves as the anchor for the nucleon mass hierarchy:

$$m_0 \approx 1.0073 \times 10^{-27} \text{ kg} \quad (2)$$

Under the principle of Failure Rigidity, any deviation in m_0 results in the structural collapse of the mass spectrum.

C. The Universal c -Anchor

Time is defined as the radial expansion property of the S^3 manifold. We reject the treatment of the speed of light (c) as a manual constant, reinterpreting it as the constant rate of time propagation:

$$r(\lambda) = \lambda = ct \quad (3)$$

This expansion forces filaments to interact with the HSUCV lattice, establishing the mechanical frequency of reality.

III. THE PROJECTION CONSTANT B

The most critical invariant for 4D-to-3D projection is the Projection Constant B . Defined as the per-dimension share of geometric distortion, it is calculated as:

$$B = \frac{3}{4\pi} \approx 0.2387 \text{ rad} \quad (4)$$

As a universal scaling operator, B governs the non-linear Gear-Box of the mass spectrum. For baryons ($Q \geq 3$), it acts as a scaling divisor, while for leptons ($Q = 1$), it serves as a suppressive fraction B_4 .

The constant B also defines the Critical Velocity threshold (v_{crit}), the point at which worldline tangents begin to precess out of the observable 3D slice:

$$v_{crit} = B \cdot c \approx 0.2387c \quad (5)$$

This constant serves as the “Geometric Fingerprint” of the lattice, ensuring that all derivations snap into place automatically.

IV. CONCLUSION OF PART II

Establishing the 24-cell HSUCV lattice and its universal anchors provides the rigid coordinate system necessary for the derivation of matter. Part III will define the Topological Grammar of matter, introducing the concepts of Topological Charge (Q) and the superhelical dynamics of particles.